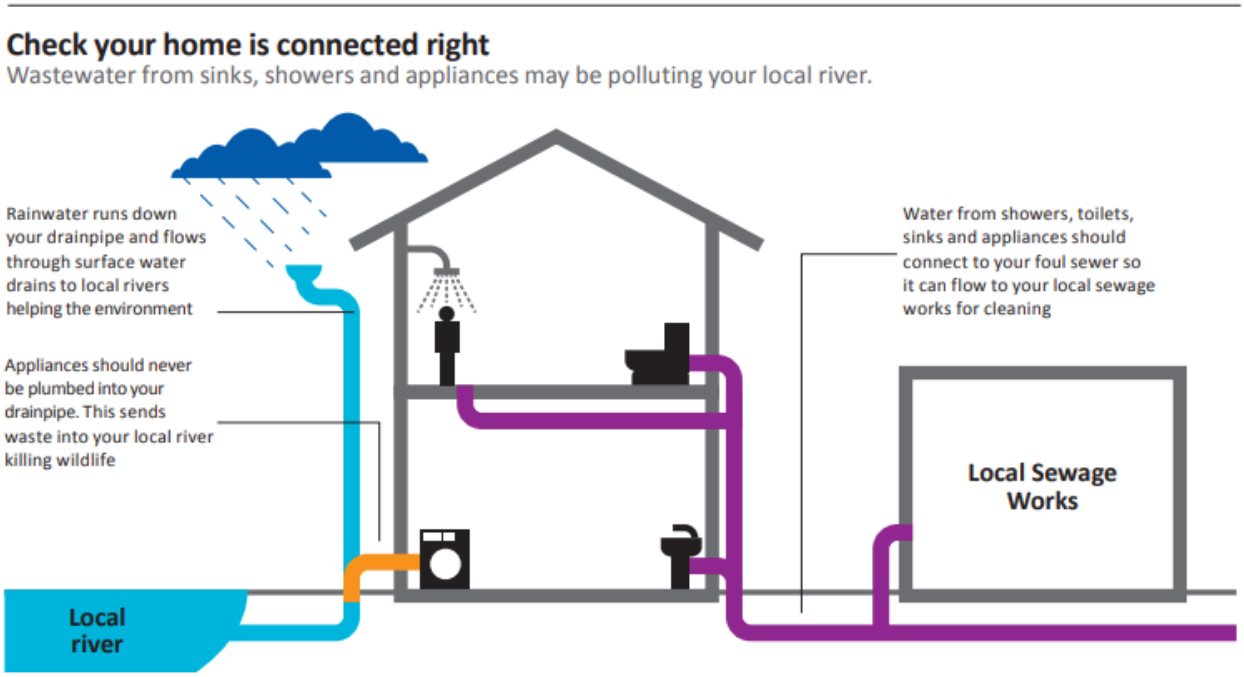
Monitoring Outfall Pollution in Birmingham

Project overview

Hi volunteers! We would like to thank you all for getting involved in this project looking at pollution from 'surface water outfalls' across Birmingham.

Surface water outfalls are pipes that discharge what should be just rainwater into rivers. This can cause pollution (e.g., from road salts and oils), but foul waste contamination in these pipes is a significant issue. This is often caused by people's foul waste appliances being illegally plumbed into the surface water sewer network, often washing machines fitted as part of extensions:



This work is part of the 'Catalyst Fund' project funded by the University's Birmingham Institute of Sustainability and Climate Action (BISCA) aimed at monitoring outfall pollution and engaging with communities on ways of addressing this problem.

We have experienced delays due to issues obtaining the equipment that will monitor pollution at high frequencies (see next steps). We have, however, made 2 important inroads:

- A law student has researched the feasibility of communities raising 'statutory nuisance notices' to help enforce outfall pollution remediation.
- We have collected 4 weeks' worth of water chemistry data from select outfalls.

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Statutory Nuisance Notices

A '**statutory nuisance**' (or abatement) notice is a legal order that holds a person/business accountable for a 'nuisance' they have created. They are normally issued by regional councils, but any 'aggrieved' person can raise a statutory nuisance against the alleged perpetrator in a magistrates' court. **But,** this requires knowing the person or business responsible. This can only be identified by Severn Trent, who cannot share details on individual properties with the public due to GDPR.

Because of this, the best way the public can help address outfall pollution is by:

- jotting down evidence of outfall pollution as a 'nuisance', including the extent of its impacts, how long it has been happening, how it has affected you.
- Record citizen science data, including Urban Riverfly, water chemistry and Urban Outfall Safari surveys as part of Birmingham River Champions.

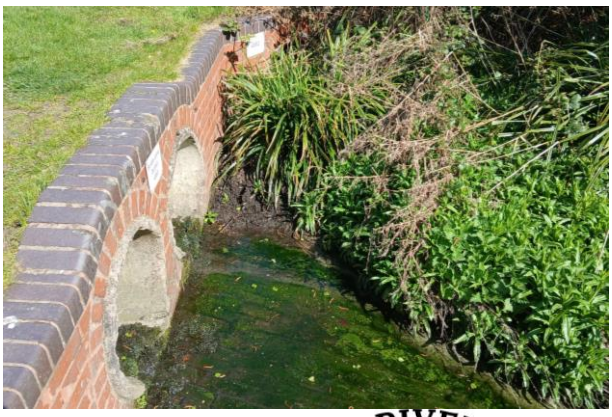
This information is valuable when reporting to Birmingham City Council's environmental health team/officer, who will then assess it and determine whether there is a nuisance. If the Council wished to pursue a statutory nuisance notice, they could then request this information from Severn Trent and start proceedings.

Water Chemistry Monitoring Sites

We focussed on 3 outfalls: **two** along Blackberry Way, Sparkhill, that have been persistently polluting the River Cole in south Birmingham (left and centre); **one** feeding a Longmoor Stream tributary in the western part of Sutton Park, near Warden's Belt (right). We sampled water chemistry within each outfall, and each river upstream (Cole) and downstream (both).







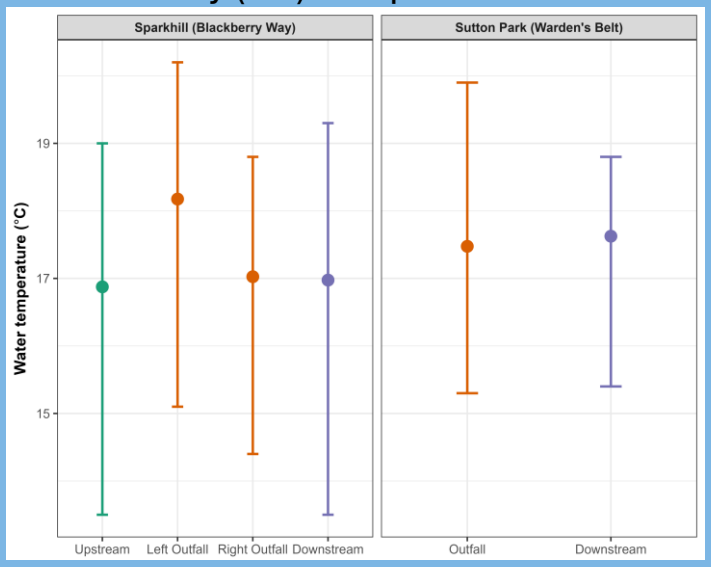
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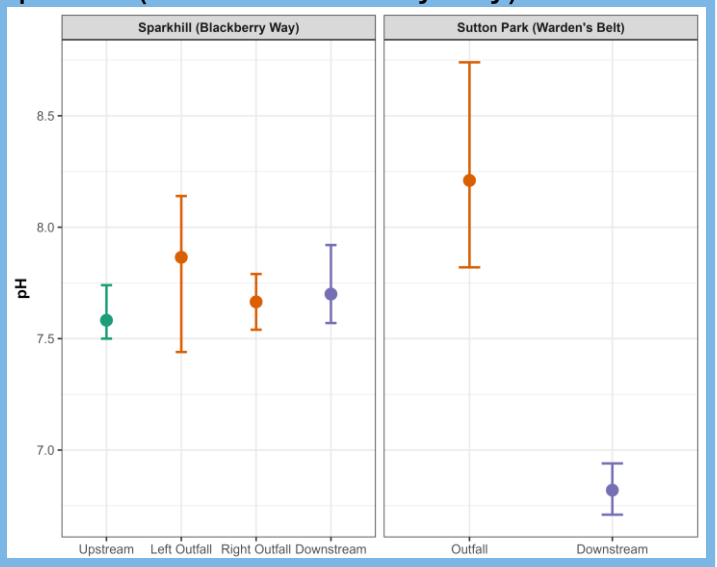
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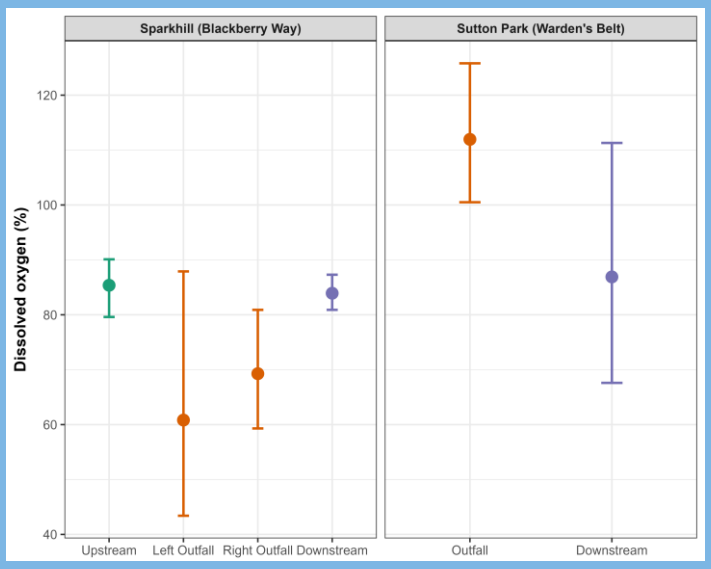
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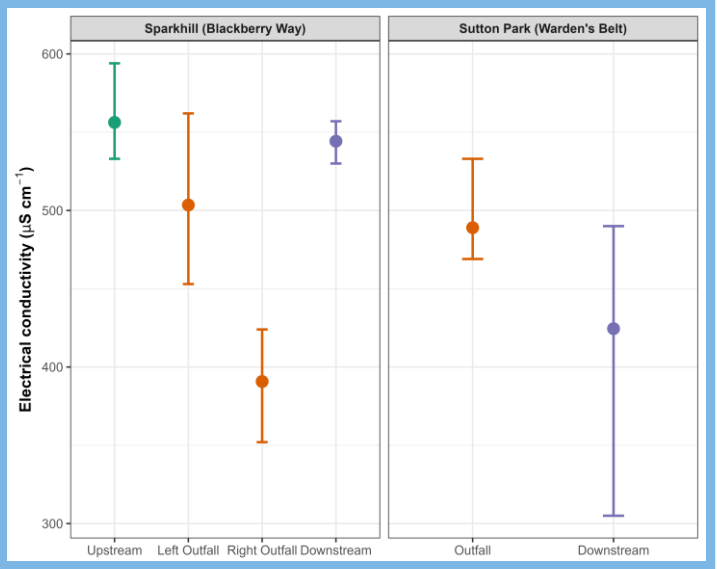
Water Chemistry Findings

So far, we have collected water temperature, dissolved oxygen, electrical conductivity (EC) and pH across a 4-week period (mid June to early July):









Both outfalls at Blackberry Way (orange) were consistently **cooler**, **more alkaline** (higher pH) and displayed **lower dissolved oxygen** compared to upstream (green). The same outfalls did, surprisingly, display **lower EC** values, which *can* indicate pollution. All factors had minor effects on downstream (purple) values. The outfall in Sutton Park had notably **higher pH** and **EC** values, and unexpectedly **higher dissolved oxygen** levels compared to downstream. Water **temperature** was largely **not affected** at Sutton Park

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Water Chemistry Caveats

The data above is preliminary and only assesses select pollution indicators. We will soon be measuring ammonia, nitrate and phosphate, which harm rivers by encouraging algae that cause overnight oxygen crashes. But we have not measured bacteria; 'sewage fungus' (actually bacteria) has been consistently present on the Cole in Sparkhill (see right).





Pollution Remediation Along the Cole

Within a separate University project called BLOSSOM, we are looking to place sacks of 'Mycelium' (part of a fungus – see left), which are excellent natural water purifiers. This concept is being explored nationally, and Anglian Water have already recorded substantial reductions of *E. Coli,* phosphate and nitrate. We are working with a consultant 'Spore and Anvill', about installing these Mycelium sacks at the site at Blackberry Way in the coming months

Next Project Steps

- We will soon provide a map of outfalls across upper Cole and Sutton Park, and invite volunteers to collect Urban Outfall Safari data.
- We are hoping to continue the water chemistry sampling indicated above.
- The high-frequency sensors are being posted out imminently and we hope to install by the end of August

The project team would again like to thank you for your interest and patience for this. If you have any questions, please email j.c.white.1@bham.ac.uk

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